

Switch Statements In Java

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What We'll Learn

By the end of this lesson, students can:

- Explain how a switch statement makes decisions.
- Use case, break, and default correctly.
- Create Java programs that select an action based on a value.
- Trace the flow of a program containing a switch statement.

Before We Start

Imagine that a program asks a user to select an option from a menu:

1. View Profile
2. Change Password
3. Log Out

How can the program display a different result for each option?

Previously, we used if, if-else, and else-if statements to make decisions. Java also provides the switch statement for choosing among several possible values.

Let's Understand

A switch statement checks the value of an expression and compares it with several possible cases.

Basic Syntax

```
switch (expression) {  
    case value1:  
        // Statements to execute  
        break;  
  
    case value2:  
        // Statements to execute  
        break;  
  
    default:  
        // Statements to execute when no case matches  
}
```

Parts Of A Switch Statement

- switch contains the value that the program will check.
- case represents one possible matching value.
- break ends the current case and exits the switch statement.
- default runs when none of the cases match.

How A Switch Statement Works

1. Java evaluates the expression inside switch.
2. The result is compared with each case.
3. When a matching case is found, its statements are executed.
4. The break statement exits the switch.
5. If no case matches, the default section is executed.

The Importance Of Break

The break statement prevents Java from continuing to the next case.

```
int choice = 1;  
  
switch (choice) {  
    case 1:  
        System.out.println("View Profile");  
        break;  
  
    case 2:  
        System.out.println("Change Password");  
        break;  
  
    default:
```

```
        System.out.println("Invalid choice");  
    }
```

Output:

[View Profile](#)

Without break, Java may continue executing the statements inside the cases that follow the matching case. This behavior is called **fall-through**.

Let's Practice

Practice 1: Trace The Program

Study the following program:

```
public class NumberChoice {  
    public static void main(String[] args) {  
        int number = 2;  
  
        switch (number) {  
            case 1:  
                System.out.println("One");  
                break;  
  
            case 2:  
                System.out.println("Two");  
                break;  
  
            case 3:  
                System.out.println("Three");  
                break;  
  
            default:  
                System.out.println("Unknown number");  
        }  
    }  
}
```

Answer the following questions:

1. What value is checked by the switch statement?
2. Which case matches the value?
3. What will the program display?
4. What is the purpose of break?
5. What would happen if the value of number were 5?

Practice 2: Complete The Switch Statement

Complete the missing parts of the program.

```
public class GradeLevel {  
    public static void main(String[] args) {  
        int gradeLevel = 11;  
  
        switch (gradeLevel) {  
            _____ 9:  
                System.out.println("Junior High School");  
                _____;  
  
            _____ 10:  
                _____;  
        }  
    }  
}
```

```

        System.out.println("Junior High School");
        _____;

    _____ 11:
        System.out.println("Senior High School");
        _____;

    _____ 12:
        System.out.println("Senior High School");
        _____;

    _____:
        System.out.println("Invalid grade level");
    }
}
}

```

Practice 3: Create A Switch Program

Create a Java program that asks the user to enter a number from 1 to 4.

Use a switch statement to display the corresponding school quarter:

- 1 displays First Quarter
- 2 displays Second Quarter
- 3 displays Third Quarter
- 4 displays Fourth Quarter
- Any other number displays Invalid quarter

Your program must include:

- A Scanner object
- A variable for the user's choice
- A switch statement
- At least four cases
- A break statement for every case
- A default section

Resources / References

- Java Development Kit
- Visual Studio Code
- Java Scanner class
- Java switch statement documentation